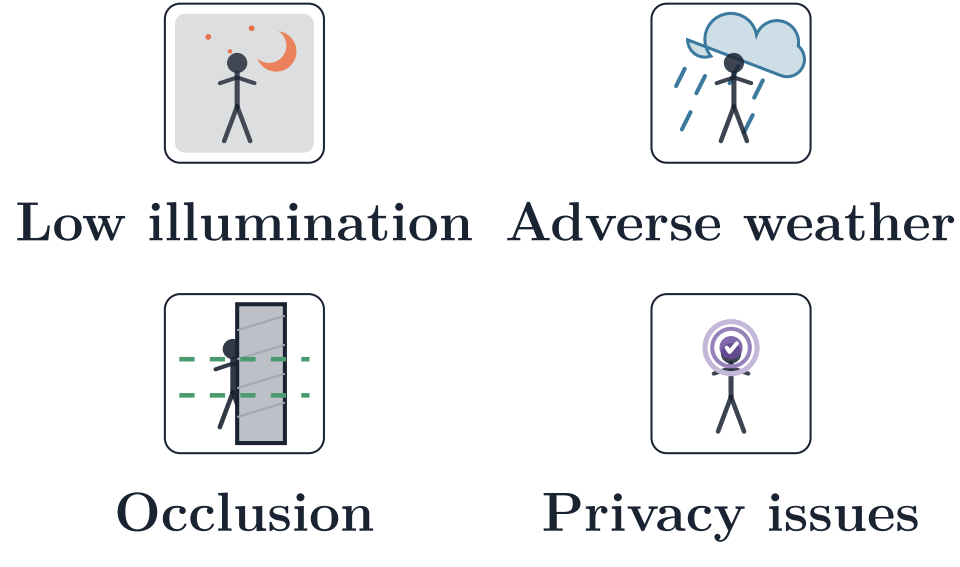


Few-Shot Radar Person Identification via Tracking-Integrated InISAR and Pre-Trained Meta-Learning

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1 Introduction

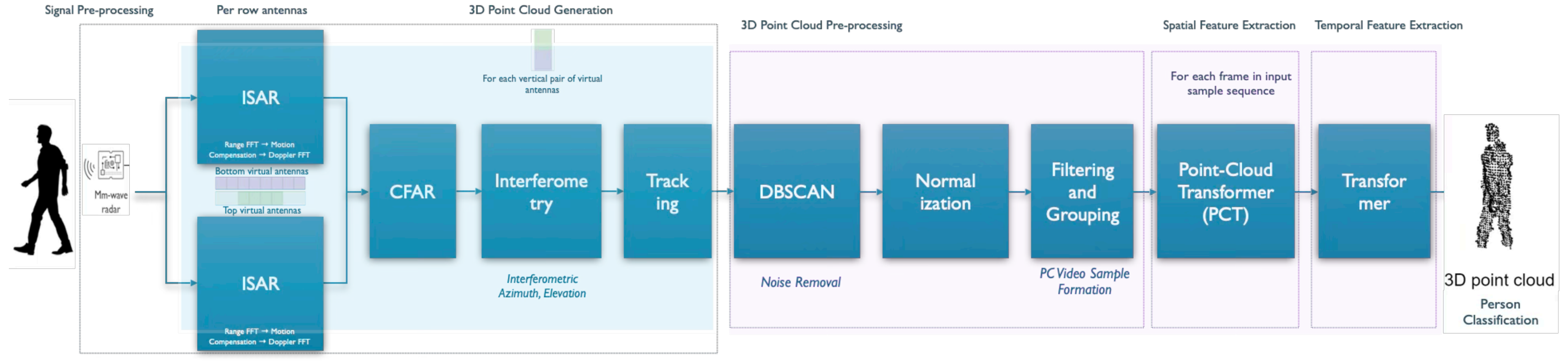
Vision-based person identification struggle under conditions [1, 2]:



Radar Advantages:

- ✓ Zero optical exposure
- ✓ All-weather operation
- ✓ Penetrates thin barriers
- ✓ Privacy preserving

4 Our Framework: Unified Sensing + Meta-Learning



End-to-end system: radar signal $\rightarrow$ InISAR 3D reconstruction $\rightarrow$ spatio-temporal learning $\rightarrow$ identity

(1) Radar domain

Tracking-integrated InISAR reconstructs high-res 3D point clouds of articulated human motion, validated against motion-capture ground truth.

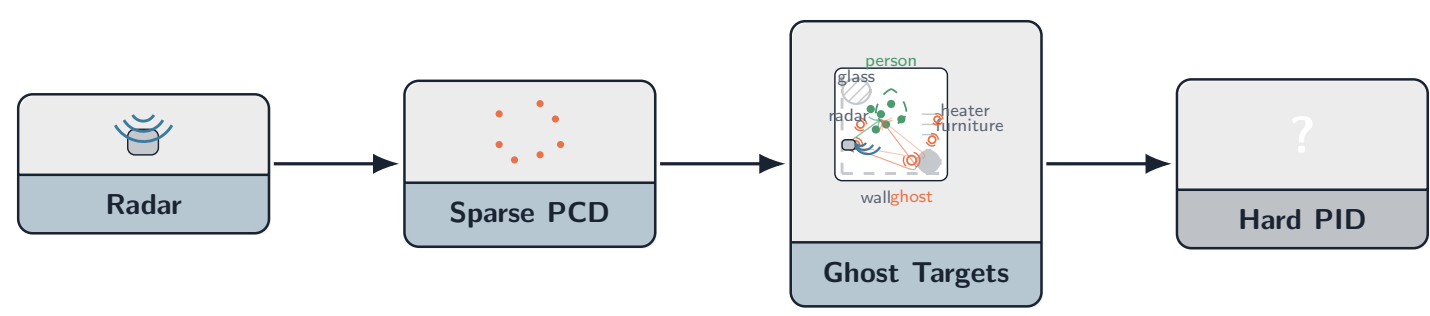
(2) ML domain

Spatio-temporal transformer (PCTrans) embedded in MAML; task-specific pre-trained initialization substantially improves few-shot adaptation.

(3) System integration

High-quality 3D reconstruction + meta-learning together enable practical few-shot radar PID, reducing dependence on large subject-specific datasets.

2 Challenges in Radar-Based PID



Radar point clouds are sparse, noisy, and contaminated by multipath ghosts

Challenge 1: Poor Point Cloud Quality

Conventional angle-of-arrival processing [3] yields sparse, noisy point clouds with limited angular resolution, motion-induced defocusing, and multipath ghost artifacts.

Challenge 2: Rigid-Target Bias in InISAR

InISAR improves angular resolution but existing works [4, 5, 6] validate it only on large *rigid* targets – **not** articulated human targets.

Challenge 3: Multipath Cannot Be Ignored

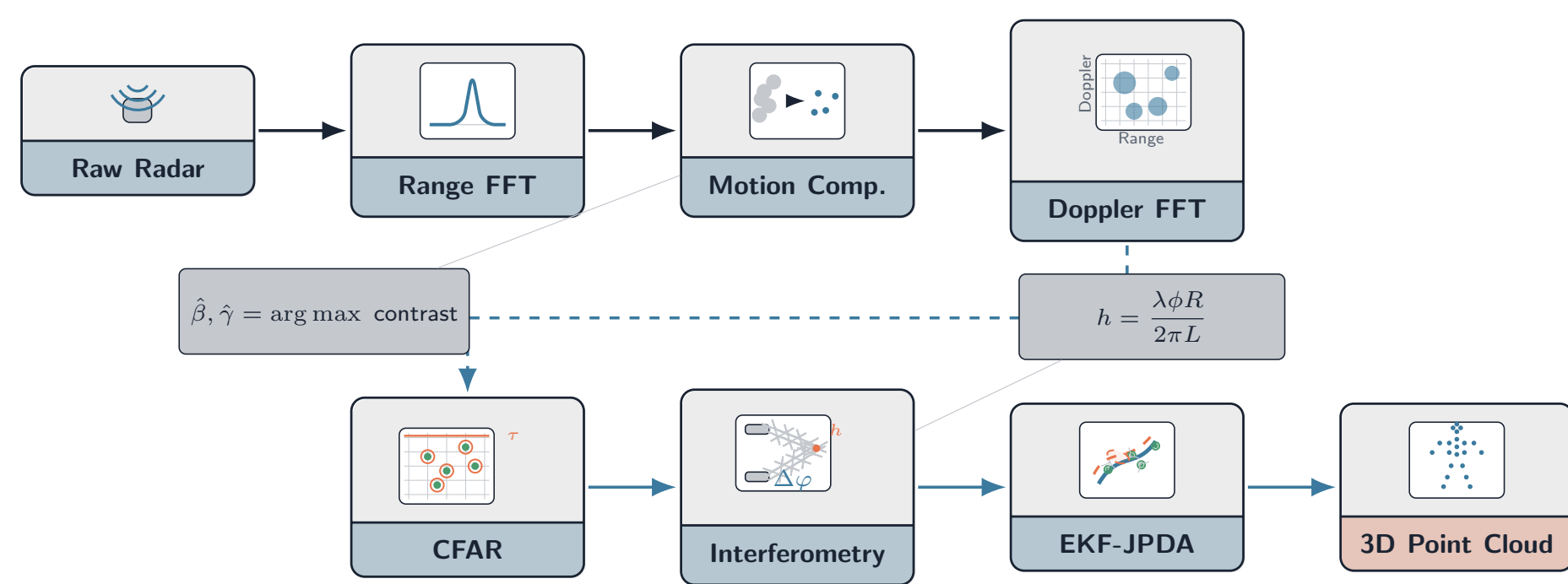
DBSCAN alone cannot distinguish true targets from multipath ghosts [7], which overlap in range-Doppler and mimic real 3D clusters.

Challenge 4: Data Scarcity

Existing methods [8, 9] need >100 sequences/person. PCT [10] handles spatial structure but targets *static* objects [11], not evolving radar sequences.

Gap: We bridge these gaps with tracking-integrated InISAR and pre-trained few-shot meta-learning.

6 InISAR Pipeline: 3D Reconstruction



Six-stage 3D reconstruction pipeline

Step 1: Range FFT

- FMCW radar transmits linear frequency-modulated chirps:

$$\Delta r = \frac{c}{2B} \quad (\text{range resolution})$$

- Output: Range-compressed signal $S_{R,m}(f, t)$ per m -th antenna.

Step 2: Motion Compensation (ICBA)

- Target motion during CPI smears energy across cellse [12].
- Translational motion model:

$$R_0(t) = r_0 + \beta t + \gamma t^2$$

- ICBA [5] maximizes image contrast to estimate $[\hat{\beta}, \hat{\gamma}]$:

$$[\hat{\beta}, \hat{\gamma}] = \arg \max \left\{ \frac{\frac{1}{N} \sum (I_i - \bar{I})^2}{\bar{I}} \right\}$$

- Output: Motion-compensated range profile $S_{C,m}(f, t)$.

Step 3: Doppler FFT (ISAR Image)

- Output: Slow-time FFT produces the 2D ISAR image (range $\times$ Doppler) with focused target signatures.

Step 4: CFAR Segmentation

- SOCA-CFAR ($N_T = 6$, $P_{FA} = 10^{-10}$) segments the target cells from noise/clutter.

Step 5: Interferometry

- Uses phase difference between top and bottom antennas:

$$I_1 \cdot I_2^* = |I_1||I_2| \exp \left(j \frac{2\pi}{\lambda} \Delta R \right)$$

$$\phi = \frac{2\pi}{\lambda} L \frac{h}{R}, \quad h = \frac{\lambda \phi R}{2\pi L}$$

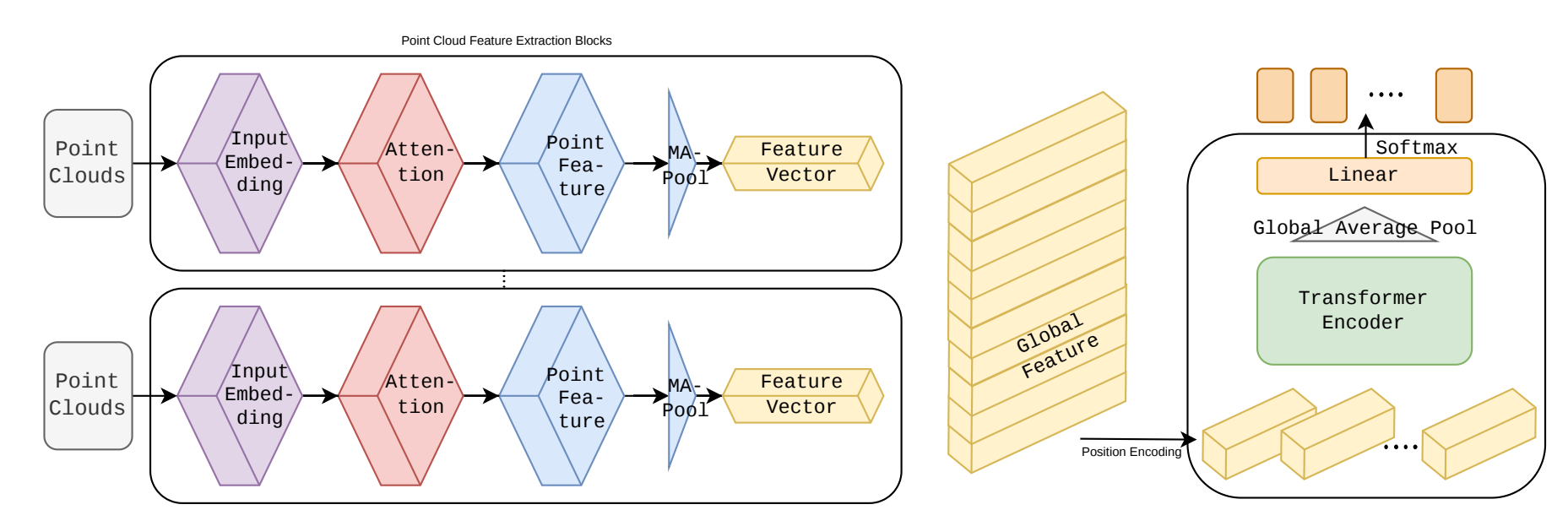
- 3D coordinates:

$$x = R \cos \theta_{el} \sin \theta_{az} \quad y = R \cos \theta_{el} \cos \theta_{az} \quad z = R \sin \theta_{el}$$

Step 6: EKF-JPDA Tracking

- JPDA handles data association; EKF filters state with covariance.
- Output: Clean 3D point cloud with ghosts removed.

7 Few-Shot PID with PCTrans



PCTrans: frame-wise PCT + Temporal Transformer

Preprocessing:

- DBSCAN ($k = 8$, $S = 1.0$) + near-field filter
- Centroid normalization: $p_i' = p_i - (\bar{x}, \bar{y}, 0)$
- Random padding to $n_{\max} = 1000$
- Output: $\mathcal{S} = \{\mathcal{P}_t\}_{t=1}^{10}$, $\mathcal{P}_t \in \mathbb{R}^{1000 \times 5}$

Spatial (PCT Encoder [10]):

- SG1: FPS $n_1 = 200$, $r_1 = 0.2$; SG2: FPS $n_2 = 100$, $r_2 = 0.3$
- Neighbor Embedding: $\Delta F(p) = \text{concat}_{q \in \text{knn}(p,k)} (F(q) - F(p))$
- Offset-Attention: $F_{\text{out}} = \text{LBR}(F_{\text{in}} - F_{\text{sa}}) + F_{\text{in}}$
- Output: $\mathbf{f}_t \in \mathbb{R}^{192}$

Temporal (Transformer):

- Positional encoding: $\mathbf{f}_t^{\text{pos}} = \mathbf{f}_t + \text{PE}_t$
- Multi-head self-attention ($h = 4$)
- Max + average pooling $\rightarrow \mathbf{g} \in \mathbb{R}^{192}$
- Classification:

$$p(y|\mathcal{S}) = \text{Softmax}(W_2 \cdot \text{ReLU}(W_1 \mathbf{g} + b_1) + b_2)$$

Key Design Choices:

- FPS + Ball Query handles non-uniform density.
- Offset-Attention emphasizes local structure.
- Temporal transformer models gait dynamics.

3 Key Results

73.8% < 5.5 cm

InISAR IoU Height Error

70.0%

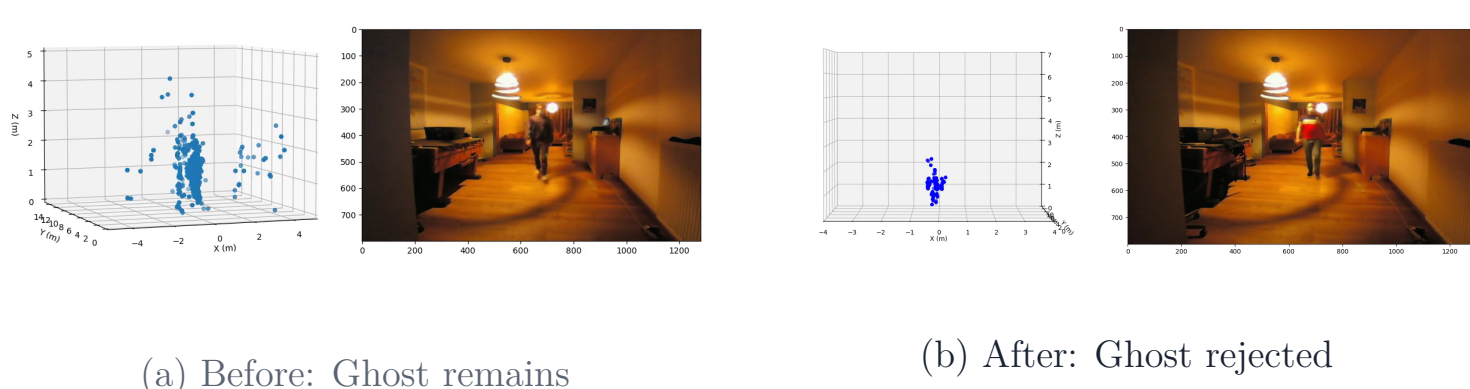
Cross-Session PID

System Impact: High-fidelity 3D reconstruction · Advanced supervised classification · Robust few-shot generalization (+15.6pp gain).

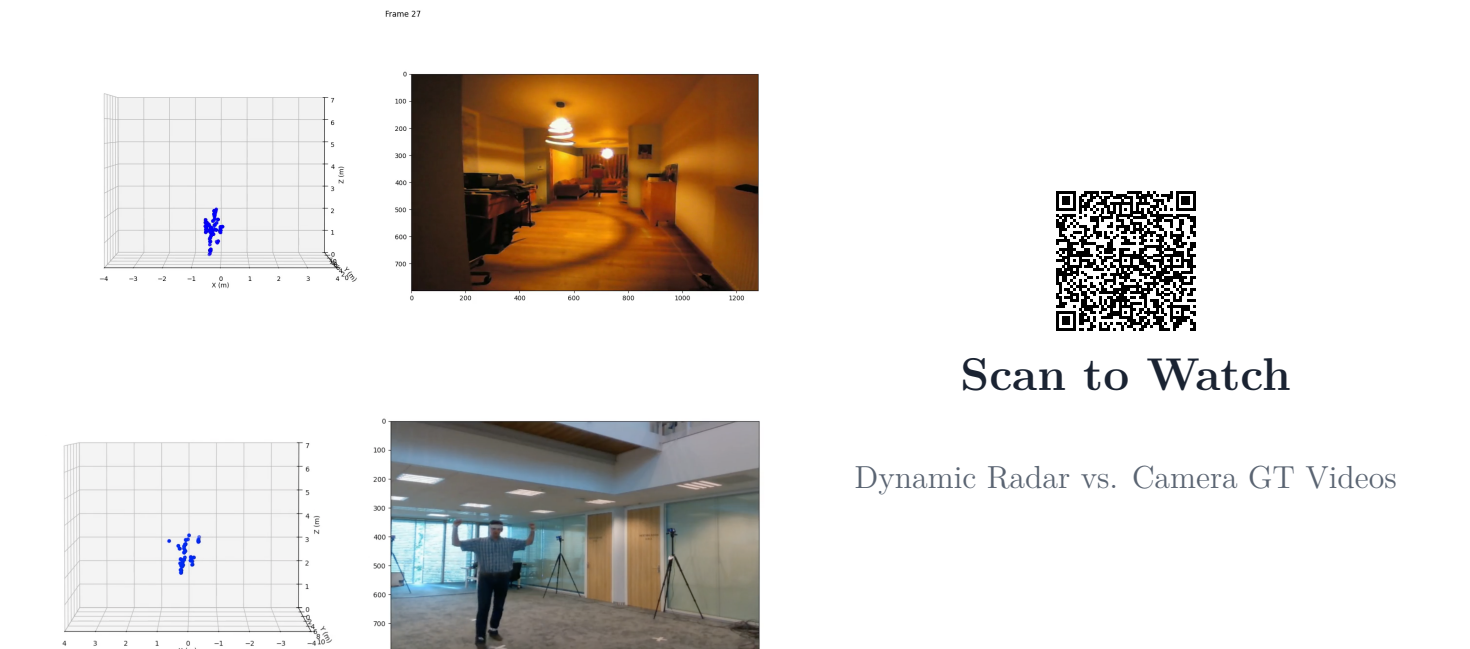
Deployable few-shot radar PID with minimal training data.

Visual & Demo

Ghost Removal (EKF-JPDA Tracking)



Real-Time Tracking Video Demo



Future Scope:

Evaluating larger population scales · Adaptive tracking for high-acceleration scenarios.

6 Validation: InISAR vs Motion Capture

Scenario	IoU (%)
Circular Walking	79.82
Hands-Up Walking	100.0
Kicking	69.19
Squatting	59.98
Average	73.78

InISAR (blue) vs MoCap (red)

Reconstruction: Height Error < 5.5 cm (median 3.87 cm)

8 Pre-trained Meta-Learning [13]

Two-Phase Algorithm:

Phase 1: Supervised training on 3 subjects $\rightarrow \theta_0^{PT}$

Phase 2: MAML [14, 15] meta-training with θ_0^{PT}

Inner loop: $\theta'_i = \theta - \alpha \nabla \mathcal{L}$ ($\alpha=0.1$). Outer loop: $\theta \leftarrow \theta - \beta \nabla \mathcal{L}$ ($\beta=0.001$).

Gain: +15.6pp over random MAML

9 Performance Evaluation & Conclusion

Benchmark Accuracy (%)

Query Type	TCPCN [9]	PCT-T MAML (sup.)	MAML (rand.)	MAML (pre-train)
Q1 (same session)	96.18	98.99	76.67	94.44
Q2 (cross-session)	35.10	32.05	54.44	70.00
Q3 (cross-subject)	57.81	40.09	46.67	48.89

- Supervised:** PCT-T outperforms TCPCN $\rightarrow$ +2.81pp (98.99%).
- Few-Shot:** Pre-trained MAML vs Random $\rightarrow$ +15.56pp (70.00%).
- mmGait Benchmark:** PCT-T leads by +11.5pp on fixed trajectories [16].